

Srinath S

AI Engineer | Agentic AI | LangGraph | MCP | Python
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Profile Summary

B.Tech Artificial Intelligence and Data Science graduate with strong CS fundamentals, DSA, and hands-on experience building agentic AI, RAG, NLP, and computer-vision systems. Built **ClosePilot**, a stateful 7-node LangGraph workflow with typed state, checkpoint-based Human-in-the-Loop (HITL) interrupts, safe pause/resume, Claude integration through MCP, structured-output validation, and a 14/14 passing Pytest suite. Also built **RepoChat**, a LangChain and FAISS retrieval system with automated response evaluation and LangSmith tracing. Portfolio & live project demos: srinath2934.github.io.

Professional Experience

Proitbridge Machine Learning Intern

Sep 2025 – Dec 2025

- Engineered a Python-based CatBoost classification pipeline using pandas and NumPy to process 100+ features from a consumer dataset, predicting loan-default risk with **90% accuracy and 0.74 ROC-AUC**.
- Optimized feature-selection and hyperparameter-tuning scripts, containerizing preprocessing and inference modules for integration testing on **AWS EC2**.
- Implemented **SHAP-based interpretability modules** in Python to explain feature contributions and communicate applicant-level predictions to business stakeholders.

Projects

ClosePilot — Multi-Agent AI Sales Copilot

[github](#)

- Built a 7-node stateful LangGraph workflow covering research, prioritization, strategy, communication, approval, action, and verification, with typed state, checkpointers, HITL interrupts, branching, and safe pause/resume.
- Integrated Claude through MCP for tool calling against a HubSpot CRM and implemented structured-output validation with read-after-write verification.
- Added LangSmith tracing for observability and verified reliability with a Pytest suite with **14/14 tests passing**.

Redrob — Intelligent Candidate Ranking Engine

[github](#)

- Built an offline, CPU-only hybrid ranking pipeline that ranked the top 100 candidates from a 100,000-candidate pool against job descriptions in **under 8 seconds of inference time**.
- Engineered a two-phase architecture with BAAI/BGE embeddings, vectorized retrieval, semantic similarity, skills, experience, and behavioral signals, producing deterministic rankings.
- Deployed an interactive Streamlit sandbox on Hugging Face Spaces with score breakdowns and CSV export for candidate filtering.

Semantic AI Ticket Routing System

[github](#)

- Architected a Python backend processing **108,819 cleaned support tickets** for classification, team assignment, priority mapping, and resolution-time ETA prediction.
- Developed Sentence-BERT embedding pipelines with FAISS for semantic search, serving predictions through FastAPI with SQLite audit logging.
- Tracked model runs with Weights & Biases; achieved **81% priority accuracy, 80% action accuracy, and 7.73-hour mean absolute error for ETAs**. Co-authored an **IEEE ICIRCA 2026 paper**.

HazWaste Detection & RepoChat (GitHub RAG Assistant)

[github](#)

- Refined and optimized a YOLOv8/v9 computer-vision model wrapped in FastAPI (latency reduced from 3–5s to 110ms) and Dockerized to shrink deployment footprint by 55%.
- Built RepoChat: a modular Python RAG assistant using LangChain and FAISS to ingest, chunk, and index 1,000+ GitHub repository files with citation-backed grounding.

Technical Skills

Machine Learning & AML – Supervised/Unsupervised Learning, Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering, Classification, Regression, CatBoost, scikit-learn, SHAP, Hyperparameter Tuning

Deep Learning & NLP – PyTorch, Natural Language Processing (NLP), Sentence-BERT (SBERT), Vectorization, Tokenization & Chunking, Text Preprocessing, YOLOv9/v8, OpenCV, Multi-Head Architectures

LLMs & Retrieval (RAG) – Large Language Models (LLMs), Claude, OpenAI, Groq LPU, RAG Pipelines, FAISS Vector Indexing, ChromaDB, BAAI/BGE Embeddings, Context Compaction, Token Budgeting, Citations

Agentic Orchestration – LangGraph (v0.2+), State Machines, Typed State, Checkpointers, Human-in-the-Loop (HITL), MCP (Model Context Protocol), Tool Calling, Multi-Agent Workflows, CrewAI (explored)

Data Science & Systems – Python 3.10+ (async, typing, OOP), SQL (PostgreSQL, SQLite, MySQL), pandas, NumPy, Data Structures & Algorithms, FastAPI, Docker, AWS EC2, LangSmith Tracing, Pytest

Education

B.Tech – Artificial Intelligence & Data Science (GPA: 8.0 / 10)

2022–2026

JCT College of Engineering and Technology, affiliated with Anna University

HSC – Mathur Boys Higher Secondary School, Tamil Nadu State Board

2022

Certifications

- Oracle Certified Foundations Associate:** OCI 2023 AI Foundations (ID: 100662362OCI23AIFCA)
- Foundation: Introduction to LangChain – Python — LangChain Academy (Jul 2026)
- Introduction to Model Context Protocol — Anthropic

- Python (Intermediate) & SQL (Basic) Certification — HackerRank
- Python for Data Science, AI & Development — IBM via Coursera
- Databases and SQL for Data Science with Python — IBM via Coursera